

SOUTHWEST TOWN MEETING

On Electricity Issues

The ways in which utilities plan for, supply, and price electricity, along with the ways users consume electricity are going through fundamental changes. El Paso Electric is asking for the suggestions of its customers to plan for these changes. El Paso Electric's service areas in Texas and New Mexico are unique. What works in other parts of the country may not necessarily be the best thing for customers in these areas.

In the next 5 years, as the economics of El Paso and Las Cruces grow, the Company foresees the need for resources in the amounts as shown in **Exhibit 5**. El Paso Electric has emerged from bankruptcy and has a 10-year rate freeze in place in Texas. Therefore, the cost of any new generation must be covered without raising base rates. There are several options. The Company wants to know about your preferences and opinions with regard to the available options

At the same time, competition is beginning to emerge in the electric industry, much like it did in the telephone industry. Competition may bring new services, new choices, and new ways to think about existing services. Competition may also bring confusion during the transition. El Paso Electric needs to know the opinions of its customers concerning new services, and the quality and method of delivery of existing services.

Why Hold A Town Meeting?

As participants in the Town Meeting, you will have an opportunity to discuss issues and options in both large and small group settings. You will also have an opportunity to ask questions of the different groups who have an interest in the outcomes of the process. After reflecting on the different perspectives and options, you should develop your own judgments based upon the things you find important. Your views as a participant in the Town Meeting are very important because, collectively, you represent the views of all 281,000 El Paso Electric customers.

This pamphlet is designed to introduce you to some of the issues and options involved with:

- selecting resources to meet future energy needs; and
- new competition and ways to deliver utility services in the future.

The large group session after lunch on Saturday will be taped for later broadcast on KCOS, Channel 13. During this televised session, you will have an opportunity to ask questions of representatives from the Public Utility Commission of Texas (PUCT). This governmental body is the state agency that is charged by state law with the protection of the public interest inherent in the rates and services of public utilities in the State of Texas.

El Paso Electric's Planning Process

State law in Texas requires each electric utility company to go through a detailed planning process as it considers the best way to meet customer needs in the future. In the law and in the rules of the PUCT, this process is called Integrated Resource Planning, commonly known as IRP.

IRP requires the utility company to consider a range of possible planning alternatives and to develop a "portfolio" or mix of resources that will provide efficient electric service in the future at a reasonable price.

The planning process is difficult for two reasons:

- no one knows what the future will bring; and
- not everyone thinks the same things are important.

Customers are the ones who pay the bills and make choices about the services they receive, so it makes sense to ask your advice when developing options to provide electric service in the future. Therefore, a part of the IRP process requires involving the public in order to gather information on local concerns about the environment, the effects of

utility operations on the local area and economy, and what customers expect from their utility in terms of cost and services. The public participation portion takes place at the beginning of the IRP process so the results can be incorporated throughout the planning cycle.

The Integrated Resource Planning Process

All Utilities:

1. Determine the need for future energy resources.
2. Gather information from the public (the step we are in now).
3. Prepare and file a Preliminary IRP with the PUCT.

Utilities Proposing A Resource Solicitation:

1. Hold a hearing on the Preliminary Plan—PUCT approves or modifies the Preliminary Plan.
2. Solicit for resources.
3. File a final IRP.
4. Hold a hearing on the final IRP to approve the contracts resulting from the solicitation.
5. Begin the process again in three years.

If the Company proposes a resource solicitation, a hearing will be held on the preliminary plan. If the Company does not propose a resource solicitation, the Preliminary Plan will be reviewed by the PUCT for compliance with the rules governing IRP.

Although El Paso Electric serves customers in New Mexico as well as Texas, the Company is not required to file an IRP in New Mexico. However, El Paso Electric plans resources for its entire system; so, it is important to involve customers from both states.

What Are Some Of The Factors To Consider?

There are several factors to consider when selecting future generation resources, including the following:

- Lowest Cost
- Environmental Impact
- Reliability
- Customer Needs
- Economic Impact
- Fuel and Resource Mix

There will be several questions that arise as you deliberate various options such as: What do you think about the costs and benefits of various options? What are your most important concerns in choosing among the options? How do you view the tradeoff between present-day costs versus costs that are delayed into the future?

Recommending the best course of action may not be easy. As with most choices, all options will have both advantages and disadvantages. What one person considers a good solution, another may not. In order to get more of a trait you like, you may have to accept more of a trait you don't like. Don't worry if you are not a technical expert. We will provide the basic information you need to make your decisions.

Traditional Fossil Fuel Resource Supply

One of the options for meeting future needs is to build a new power plant. Traditional resources such as coal and natural gas have a long history of development, and their efficiency and performance continue to improve.

Purchased Power

A different way to think about meeting future generation needs is to purchase power in the open market without regard to fuel. Federal rules mandate open access to other electric utilities' transmission systems, allowing access to more sources of energy. As a result, active markets that buy and sell electric power have developed an effective approach for acquiring power for many companies. Purchased power contracts can be for long-term or short-term and can have fixed prices or prices that float with changing events such as fuel cost or financial markets. The contracts are based on the needs of buyers and resources of sellers. The use of



purchased power may allow a utility to avoid the construction of new generation.

Efficiency Programs

A third option for meeting future energy needs is to develop conservation programs to reduce the energy use of individual customers and avoid the need for a new plant. Because El Paso Electric's energy use per person is relatively low, it is unlikely this approach could satisfy all of the future need, but it might satisfy part of the resource needs. Energy efficiency means more than just not using electricity. It often requires investments to make buildings and factories more energy efficient. In some cases, energy efficiency is less expensive and, in other cases, it is more expensive than other options. Actions taken by customers can shift usage from one time period to another, thereby reducing the overall cost to the utility. Load management programs can benefit the utility system by reducing peak usage or by increasing the use of more efficient generating units.

Renewable Energy

A fourth option is to build facilities that use renewable energy sources. The costs associated with building renewable energy facilities are still more expensive than traditional power sources, but costs are beginning to come down. The most common renewable sources are wind and solar energy, which have no fuel costs. Other renewable resources which have some fuel costs include methane gas developed from landfills or agricultural products. In general, renewable energy is seen as providing strong environmental benefits.

Competition And New Ways To Deliver Utility Services

A special topic you will consider is competition in the electric industry. In the future, you will likely have a choice of providers for electricity and a broader range of services. This topic was debated during the 1997 Texas and New Mexico legislative sessions, and is the subject of ongoing discussions at both state commissions. At this time, no one knows exactly how the industry will change, only that it will be different in coming years. It is this period of transition to a competitive environment that provides a high level of uncertainty for El Paso Electric's planning process. The Company does not expect you, as customers, to predict what will happen concerning future competition, but is interested in what you think about existing services and how those services could be improved in the future. El Paso Electric wants to outline some possible new services and get your reactions. This part of the Town Meeting will require some imagination in thinking about the future.

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About These Materials

These materials were prepared with the invaluable assistance of an Advisory Group comprised of people with diverse viewpoints on the issues discussed. Their contribution was to see that the discussions are balanced and that all viewpoints are fairly presented.

Background Information on El Paso Electric

Introduction

Customers are affected by energy resource decisions because they pay the bills, but not all customers are affected in the same way or have the same interests. Likewise, there are a number of other groups who participate in energy resource decisions or are affected by the outcome of these decisions. These other groups include shareholders who own El Paso Electric stock, regulators who set rates and service standards, companies and individuals who supply services and products to El Paso Electric, emerging competitors who want an opportunity to provide the same or similar services, and customers who can decide to generate their own electricity by building generation facilities. Therefore, one of the major challenges of the planning process is to find a set of solutions that fairly balances the interests of these different groups.

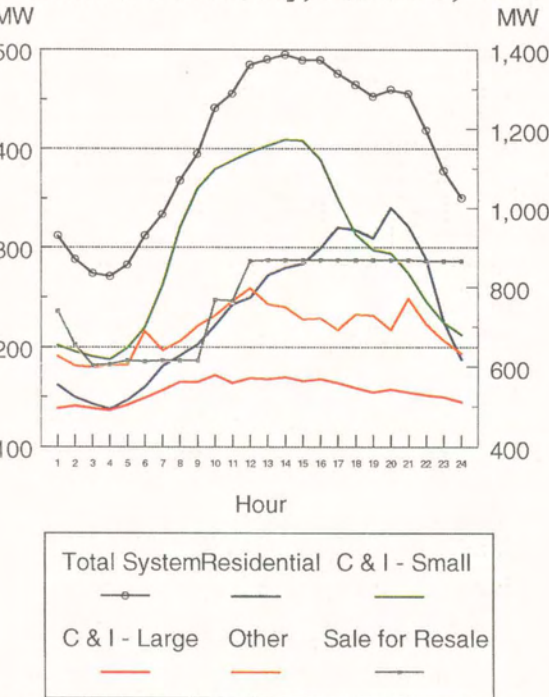
El Paso Electric is a regulated monopoly. In exchange for the exclusive right to provide electricity within its service territory (see Exhibit 1), El Paso Electric has an obligation to serve all customers who wish to purchase the Company's services. El Paso Electric's status as a regulated monopoly means that the Company must follow the rules and procedures set by the PUCT and the New Mexico Public Utility Commission (NMPUC). The City of El Paso, along with other municipalities, also regulates El Paso Electric within the city limits, and interstate transactions are regulated by the Federal Energy Regulatory Commission (FERC). Governmental regulators, like groups of customers, also sometimes have differing interests and priorities. These interests and priorities must also be factored into the planning process.

Types Of Customers And Their Concerns

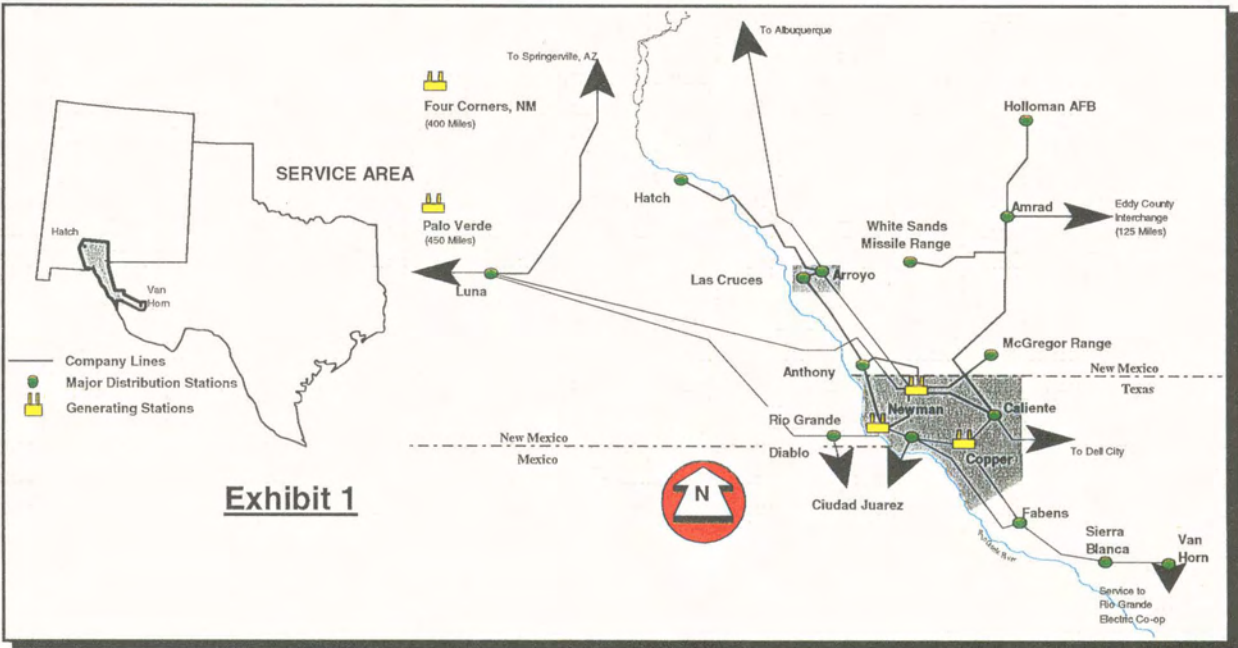
El Paso Electric has three main groups of "retail" customers: residential, commercial, and industrial. Participants in the Town Meeting will represent mostly residential customers, but there will also be a sampling of small commercial and industrial customers. El Paso Electric also serves specialized types of loads such as military bases, universities, and large commercial and industrial customers that are capable of generating their own electricity. Exhibit 2 illustrates electric load by customer type. In addition, El Paso Electric makes wholesale sales to other entities who, in turn, resell the electricity to their customers.

Exhibit 2

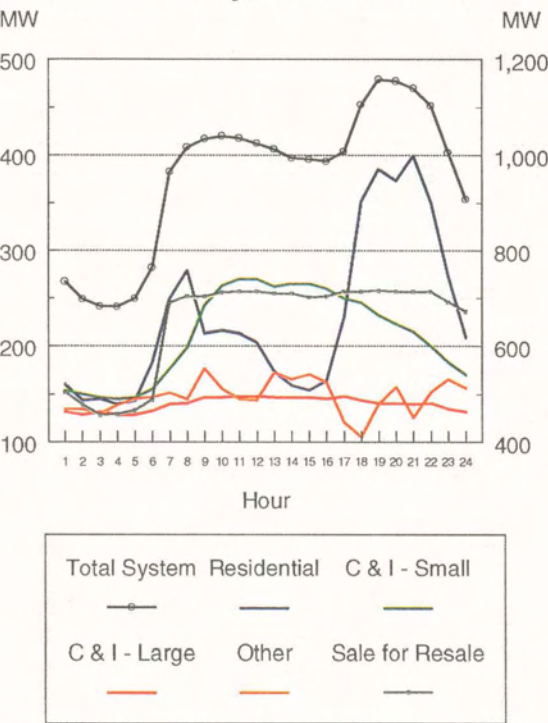
Summer Peak Day, June 20, 1996



Total System uses Right Scale, all others use Left
Other includes Other Public Authorities and
Sale for Resale includes CFE, IID, TNP and RGEC.



Winter Peak Day, December 17, 1996



Total System uses Right Scale, all others use Left Scale

Residential Customers

El Paso Electric has 194,000 residential customers in Texas and 57,000 residential customers in New Mexico. The average Texas/New Mexico residential customer uses 520 kWh of electricity and pays approximately \$54 a month when averaged over the entire year (A kWh is 1,000 watts used for an hour—think of 10 light bulbs each using 100 watts, burning for an hour). While El Paso Electric's average residential energy use is the lowest in Texas, there can be dramatic differences in residential bills based on customer choices such as the use of refrigerated air conditioning. Exhibit 3 shows typical residential usage by percentage.

Exhibit 3

Typical Residential Usage by Percent	
with Evaporative Air Conditioning	
Annual Energy Usage, kWh	6,188
Monthly Average Usage, kWh	516
Total	100%
Refrigerator	28%
Clothes Dryer	11%
Evaporative Air Conditioner	7%
Lighting and Other Appliances	54%
with Refrigerated Air Conditioning	
Annual Energy Usage, kWh	8,724
Monthly Average Usage, kWh	727
Total	100%
Refrigerator	20%
Clothes Dryer	8%
Refrigerated Air Conditioner	29%
Space Heating, Lighting and Other Appliances	43%

Note:
1. The difference between these two typical patterns of usage reflects an average monthly increase of 211 kWh for refrigerated AC over evaporative cooling (or 422 kWh averaged over the six cooling months).

Commercial And Industrial (Small)

This group includes businesses such as supermarkets and other types of stores, restaurants, office buildings, schools, churches, and small manufacturing facilities. The amount of energy these groups use can differ greatly depending upon the size of the building, the type of heating and cooling equipment they use, and any special equipment such as freezers, cooking equipment, motors, and fabrication tools such as welders. El Paso Electric has 18,000 small commercial and industrial customers in Texas and 6,000 small commercial and industrial customers in New Mexico.

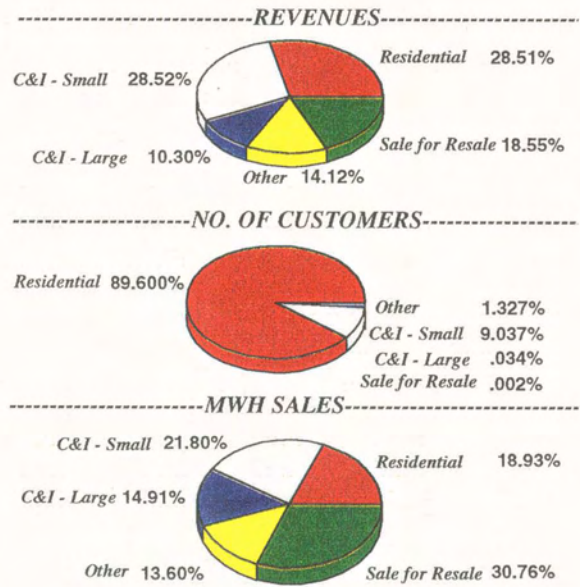
Commercial And Industrial (Large)

This group includes the largest energy users on the El Paso Electric system. There are 91 large commercial and industrial customers in Texas and 7 large commercial and industrial customers in New Mexico. Refineries and steel companies are in this category. These customers use large amounts of electricity. It is not uncommon for these customers to operate 24 hours a day.

Each group of customers has particular needs and patterns of using electricity. Therefore, different types of customers might prefer different options for electric service. For example, residential customers tend to use more electricity from 7 p.m. to 8 p.m. each day. Industrial customer usage tends to be more constant. Exhibit 4 depicts percentages by group for revenues, customers and sales.

Exhibit 4

EL PASO ELECTRIC COMPANY
STATISTICS FOR 1996
(Includes CFE)



Planning With Uncertainty

El Paso Electric's planning for future needs has always included consideration of the uncertainties of forecasted versus actual customer growth and the demand for power, and the ability of large customers to generate their own electricity. These uncertainties persist. Additionally, El Paso Electric's plans for future needs are affected by the current uncertainties associated with the future of the Las Cruces load and the future of the Company's wholesale contracts for power sales outside the El Paso Electric service territory.

Exhibit 5

El Paso Electric Company's
Projected Need For Additional Resources

Year	Amount (MW)
1998	50
1999	75
2000	100
2001	125
2002	75

Las Cruces Issues

The City of Las Cruces has voted to purchase the assets of El Paso Electric and to form its own electricity utility. Legislation that would permit Las Cruces to condemn these assets was passed in the 1997 Session of the New Mexico Legislature. El Paso Electric opposes a forced taking of its property and believes that the cost Las Cruces would have to pay for such property would ultimately require Las Cruces to establish rates at higher levels than currently charged by El Paso Electric. The issue remains highly contested and is the subject of several actions before New Mexico state and federal courts. The outcome of this contested issue will impact the future of El Paso Electric.

Wholesale Sales

El Paso Electric has long been a supplier to the Comision Federal de Electricidad (CFE), which serves the City of Juarez, in Mexico. El Paso Electric also has been a long-time supplier to Texas-NM Power, Public Service Company of New Mexico,

Rio Grande Electric Co-Ops and the Imperial Irrigation District in Southern California. While these wholesale transactions are important sources of revenue to support the electric system operated by El Paso Electric, some regulators believe that small retail customers will be treated more fairly if there is a clear separation between the regulated retail operations of the utility and the competitive wholesale activities.

Bankruptcy And Shareholder Issues

In January 1992, El Paso Electric filed for protection under U.S. Bankruptcy Law under Chapter 11 of the Bankruptcy Code to reorganize its financial obligations. El Paso Electric reached an agreement with its creditors and filed a reorganization plan approved by the Bankruptcy Court, which became effective on February 12, 1996 and which allowed El Paso Electric to emerge from bankruptcy.

The financial condition of El Paso Electric, having recently emerged from bankruptcy, is still delicate. The reorganization plan, which resulted in a 10-year rate freeze for customers in Texas, must be considered during the planning process.

Interests Of Regulators

El Paso Electric has four main groups of utility regulators. These include the City of El Paso (and other Texas municipalities that have retained jurisdiction), the PUCT, the NMPUC, and the FERC. Each of these governmental bodies sets rates and service quality standards and approves transactions. With leadership from the City of El Paso, the Company in 1995 signed an agreement, including a

10-year rate freeze, that was found by the PUCT to be in the public interest. In addition, El Paso Electric is participating in a collaborative effort with the NMPUC to investigate the impact of possible competition on electric service in New Mexico.

Participants in the Town Meeting will have an opportunity to discuss their views with and ask questions of Texas regulators during a session on Saturday evening to be taped for later broadcast over Channel 13, KCOS.

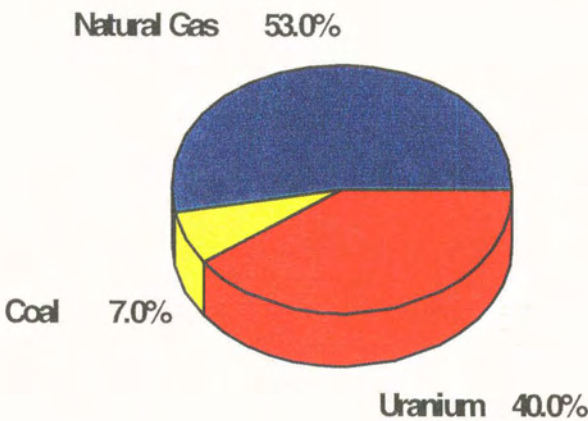
Other Types Of Interested Groups

A number of other groups are interested in how El Paso Electric plans to meet its future energy resource needs. These groups include those with environmental interests, those with consumer interests, and those representing low-income customers. In addition, there are large industrial consumer organizations, state agencies, universities, and city governments, who are interested in El Paso Electric's plans.

Additionally, the Texas Office of Public Utility Counsel (OPUC) is a state agency that was established in 1983 to provide representation for residential and small business customers. OPUC participates in major rate cases and rulemaking proceedings before the PUCT. In New Mexico, the Office of Attorney General serves a similar function as a representative for residential consumer interests.

Exhibit 6

EPE'S EXISTING GENERATION
CAPACITY BY FUEL TYPE



EPE'S 1996 ELECTRICITY PRODUCED
BY RESOURCE TYPE

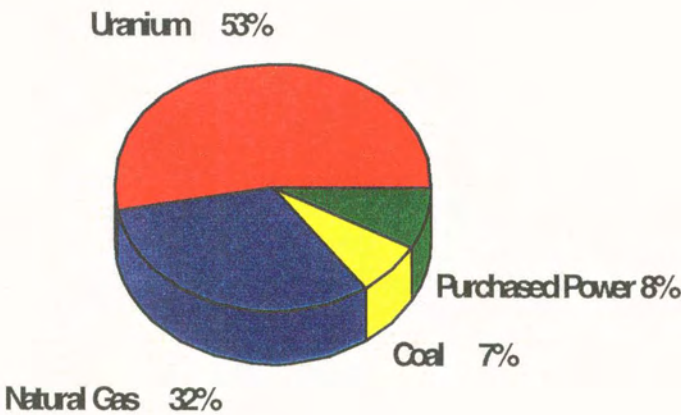


Exhibit 7

EXISTING GENERATION SOURCES								
Generating Stations (1)	Size, MW (2)	Fuel Type	Remaining Life of Units, Years (3)	Capacity Factor, % (4)	Plant Costs, \$/kw	Operational Costs		
						O&M, ¢/kWh	Fuel, ¢/kWh	Total, ¢/kWh
Palo Verde	600	Uranium	29-31	86	2386	1.1	.5	1.6
Four Corners	104	Coal	22-23	66	600	.5	1.4	1.9
Newman	482	Natural Gas	13-28	44	167	.5	2.4	2.9
Rio Grande	246	Natural Gas	10-25	38	135	.9	2.7	3.6
Copper	68	Natural Gas	23	15	211	.6	3.6	4.2
Purchased Power	Dependent on available transmission capacity and market conditions.	Controlled by market conditions.	Controlled by market conditions.	NA	(5)	NA	2.7	2.7
Total	1500	-	-	-	-	-	-	-

Notes: 1996 Data
(1) Station site may contain more than one generating unit.
(2) EPE's share of Palo Verde is 600 MW and EPE's share of Four Corners is 104 MW. EPE's owns 100% of the other generating units.
(3) Remaining life varies according to age of units on site.
(4) Capacity Factor is the percentage of maximum energy the unit delivered. It accounts for maintenance outages and economic down-time.
(5) Although there are no "Plant Costs" per se for purchased power costs, these costs may include a demand charge.
NA-Not Applicable

Factors to Consider

Your Role

No one else can represent your unique set of experiences, concerns, or individual priorities. This part of the deliberation centers on factors you may want to consider in making recommendations about future sources of generation, and programs and services you believe the utility company should offer. Some of these factors may be more important to you than others – this is what we want to know.

You will consider these factors in the first small group session on Friday afternoon. Think of the factors as different ways to view the issue of providing electric service in the future. Little discussion is necessary to realize that people differ in their views of what is important about electric service. There are no right and wrong answers. El Paso Electric does not seek a group consensus or group answer. Use the discussion and the chance to ask questions as a way to sharpen your own thinking about what you believe is important. There may be factors other than the ones listed here that you wish to discuss with your fellow small group members.

As a group, you will develop one or two questions you wish to ask the first large group panel. Ask whatever you want; this is your deliberation. Some of the most useful questions probe experts' opinions concerning tradeoffs between priorities or advantages and disadvantages of a particular course. Do not be afraid to put the experts on the spot or ask follow-up questions. The large group moderator will help.

It is not uncommon to have more questions than time. It is also not uncommon for several groups to ask similar questions. For that reason, we combine questions and, on occasion, not all questions get answered. But, over the course of the 1-1/2 day deliberation, we try to see that each group has a fair number of its questions asked in the large group sessions. For each question submitted by a small group, the members should select a person from the group to pose the question. If the question is selected, that person from the small group rises, is handed a microphone, and reads the question. The moderator then directs the questions to specific experts on the panel and sometimes allows additional experts to comment should they have differing views. As stated earlier, follow-up questions are permitted. Remember, this is your deliberation. This is not a time for speech-making by experts. We want you to receive the answers you seek and at the end of the deliberation, we want to know what you think as an individual.

Lowest Cost

The cost of electricity is determined by a combination of factors. In general, these include:

- construction cost or "capital investment"
- operation and maintenance expense
- fuel expense (fuel is a major cost of electricity production today)

Some of the ways to think about costs include short-term versus long-term costs, and the cost to build versus the cost to operate. As you will see in later discussions, some options have high upfront costs but low operations, maintenance and fuel costs. Others have low upfront costs and higher fuel costs.

Some customers, especially those facing stiff competition in their businesses, or those on fixed incomes or low incomes, may believe the most important factor in selecting a resource should be cost. They say that while other priorities might be "nice to have," they prefer not to pay the added costs. In the short-term versus long-term cost debate, they argue they must first survive today in order to worry about tomorrow. Their preference is that new generation resources be put out for bid and the lowest bid selected, regardless of other concerns.

Another viewpoint on costs is the desire for stability. Some customers are willing to pay a little more upfront if the long-term operation and fuel costs of a particular generation method are stable and predictable. This group of customers does not like surprises in dealing with expenses and gives stability a high priority. As an example, in financing a home this group of customers might pick a fixed-rate, long-term mortgage over a variable rate mortgage which initially has a lower interest rate, but changes over time.

Environmental Impact

Any electrical generation process that burns a fossil fuel creates emissions commonly known as pollution. Different plants have different emission levels. Some of the emissions are controlled by national standards and others are not. Any electric plant must meet all current state and federal environmental standards. However, mere compliance with state and federal laws does not eliminate all pollution. The Company can make additional pollution control investments. One unknown is that standards may be higher in the future if environmental laws change or environmental problems increase. The major pollutants are as follows:

- Oxides of Nitrogen—components of smog and ozone.
- Sulfur Dioxide—leads to acid rain.
- Carbon Dioxide—greenhouse gas that may lead to climate change and is not currently controlled.
- Fine Particles—lead to haze and can affect breathing.

Power plants, even when they are in full compliance with all environmental laws and standards, are a source of emissions, much like automobiles. The El Paso Electric plants are in full compliance with all federal, state, and local laws and regulations regarding emissions standards, but still produce emissions as shown in **Exhibit 8**.

Some advocates believe a utility should consider possible cleanup costs and health costs to residents when selecting resources and determining the "lowest resource cost." In addition, they are concerned about rising costs of environmental controls. Finally, environmental advocates are concerned about impacts on the climate due to greenhouse gas emissions.

El Paso is the only city in Texas in violation of national standards for both respirable particulate matter and carbon monoxide. The City is also designated as a "serious" non-attainment area for ozone; that is, it has not been able to meet federal air quality standards. However, El Paso has been in compliance with national standards for nitrogen dioxide, sulfur dioxide, and since 1986, levels of lead.

El Paso and other cities along the Texas/Mexico border face unique pollution problems because environmental standards differ on the two sides of the border. Joint efforts between the U.S. and Mexico show great promise for cooperative solutions to common environmental problems.

Reliability

In addition to normal small area outages, El Paso Electric suffered four system-wide outages in 1995 and 1996. Outages can be caused by a failure of generation plants or failures of the transmission and distribution system. Different types of generation have different reliability records. Power delivered from long distances faces greater risk of transmission failure.

El Paso Electric and other utilities are required by law to provide their customers with a reliable supply of electricity. Because the technology to store electricity is limited at this time, electricity must be generated at the time customers want to use it. There is one exception. A large C/I customer may voluntarily allow the interruption of its power supplies under certain conditions in exchange for a lower rate. These customers are called "interruptible customers."

A high level of reliability is maintained by the Company through several measures. National reliability councils set standards for lowering the risks of outages. Utilities in the Western Systems Coordinating Council, of which El Paso Electric is a member, maintain a reserve margin of capacity equal to the largest possible element of electricity production which might fail, plus 5% of peak load demand. The Palo Verde Nuclear Power Station is El Paso Electric's largest source of electricity production. Because Palo Verde generates a large amount of power, El Paso Electric must have a large margin of reserve to back it up.

Reliability standards are necessary because no source of power can be available all the time. Even though some plants, like Palo Verde, run continuously, these plants must be shut down

periodically for scheduled maintenance. They may also be out of service because of unforeseen operating problems or bad weather.

Diversity of resources is critical to the reliability of an electric utility system. The idea is to "put your eggs in more than one basket" to minimize risk. Small plants are better suited for reliability than large plants. If a small plant fails, the shortfall can be more easily covered.

Another aspect of reliability is the operation of the transmission and distribution systems. Transmission systems move large quantities of power, over long distances, to local distribution systems, which transform the power flows into smaller quantities for delivery to your home.

One way to think about reliability is in terms of risk, much like you do when purchasing insurance. You can over-build or build duplicate lines to lessen risk, but there is a cost. Since the system-wide outages that occurred in September and October '95 and March '96, El Paso Electric has devoted significant resources to ensure, to the greatest extent reasonably possible, that these types of outages do not occur again.

Toward that end, the Company has enlisted the expert advice and services of such industry leaders as General Electric, Bonneville Power Administration, the Electric Power Research Institute, and the Western Systems Coordinating Council. In addition, the Company convened a 10-person panel of academic and industry experts to assess the conditions that led to the outages and provide remediation assistance. Based upon the analysis performed by the Company and its advisors, the Company has undertaken considerable steps to address the events that led to the outages.

El Paso Electric has significantly strengthened its power supply system by:

- greatly reducing faults on a major import transmission line
- substantially modifying its transmission relay system to prevent misoperations
- implementing remedial action schemes that decrease the likelihood of system collapse during emergency situations
- increasing training to better prepare operators to deal with such emergencies and therefore hasten restoration efforts

While no utility can guarantee that system-wide outages will never occur, the Company is confident that its corrective efforts to date have been and will continue to be effective in significantly reducing the likelihood that system-wide outages will occur again in the future.

Meeting Basic Needs Of All Customers

Another factor to consider is how an option for generation resources or a new utility program contributes toward the goal of meeting basic needs for all customers. State law requires that all customer classes be treated equitably. Still, some customers have difficulty paying for utility services.

It is possible to structure programs that help customers meet their basic needs. Advantages of these programs are both humanitarian and economic. The humanitarian benefits of such programs are obvious. Those who argue the economic benefits say that programs providing assistance for basic needs are a good long-term investment. Such programs allow those in need to transition from assistance to economic contribution.

Current El Paso Electric programs that help customers meet basic needs include LIFT, Low-Income efficiency Tools. The goal is to assist low-income families in lowering their energy consumption and subsequently their energy costs. They thereby also increase their comfort levels, health and well-being. This is done by weatherizing their homes.

LIFT is complemented by Project Bravo, offered by the Texas Department of Housing and Community Affairs. For families who do not exceed 125% of the federal poverty level, Project Bravo provides caulking, weather stripping, insulation and structural repairs. Limited amounts of energy assistance are also available from other government and privately-funded programs.

Economic Impact

Any decision to add resources will have some effect on employment and business growth. Should El Paso Electric take these kinds of effects into account in its evaluation of resources? Some believe that building fossil fuel plants would be a boost to the area's economy through additional jobs and an increase in the tax base. Others believe that stimulation of the energy efficiency and renewable energy industries is best for the economy. In the long-term, they say, these resources will help to make the region more energy independent, add new industries and create more new jobs for the region than a fossil fuel plant.

Others believe that the least expensive electric service will attract business, thereby boosting the economy. They cite the need to have sufficient and low cost power available. Still, others want to make sure new power sources will not trigger new environmental requirements that limit industrial expansion. Some customers do not believe public utilities should be incurring costs to engage in economic or social policy. Yet, all business operations, no matter their size, reflect both an economic and social focus. The idea of business decisions being made that lack either consideration is unrealistic. Your deliberation can help El Paso Electric assess a proper balance.

Certain El Paso Electric customers are big enough to generate their own electricity. These might include a university, refinery or military base. Some small customers worry about big customers looking

after themselves and leaving the system, thus increasing costs for the remaining customers. Others want large employers to remain economically healthy and encourage them to find ways to cut costs. Clearly, the economic impact of any resource decision will have far-reaching implications for the El Paso economy.

Fuel And Resource Mix

An electric utility's mix of resources is critical to maintaining system reliability. Once again, diversity of fuel type, not having all your eggs in one basket, is beneficial. This diversity can be achieved by balancing the fuels used for generation. In addition to size, diversity can be achieved in power markets by acquiring power contracts with different lengths (terms). Fuel and power purchases can be short-term (priced at the "going rate") or long-term (priced at a negotiated price, a price index or futures price market). One way to address diversity is to purchase from a variety of sources.

El Paso Electric's fuel mix is currently made up of natural gas, coal, and uranium. The coal and uranium suppliers have predicted that supplies will be adequate for the foreseeable future and are willing to sign long-term contracts in exchange for more stable prices than those in the spot market. Natural gas supplies, on the other hand, are more volatile. Historically, the marketplace for natural gas has seen spikes in prices and, at times, disruptions in fuel supply. With deregulation of the natural gas market, the industry is rapidly changing. New market mechanisms provide opportunities for risk takers and

more stability for those who prefer fixed or indexed prices.

Some consumer and environmental groups believe that the best way to hedge bets on fuel availability and fuel price is to diversify the energy mix with energy efficiency and renewable resources. Energy efficiency and renewable resources are fuel-free and are therefore not subject to market price swings. As with any commodity or service market, the availability of supply and the level of demand may impact costs of renewable energy sources. Some note that solar and wind energy are dependent on natural forces, such as weather, and that care must be taken to place (site) a unit properly. In the marketplace, persons with expertise in such technologies can assume the financial risks of their decisions, and consumers can be largely protected from such risks.

Energy For Future Generations

Given the debate over the supply of natural gas, some customers are concerned about having sufficient fuel available for future generations. They believe more emphasis should be placed on energy sources that are not constrained in supply and on the efficient use of existing supply. These generating sources include solar, wind, or other renewables, such as biomass fuels, which include landfill gas. Efficient use of existing supply relies on more energy-efficient homes, appliances, and industrial processes (motors and machines).

Exhibit 8

EPE'S EXISTING GENERATION SOURCES					
Plant Sites (1)	Size MW	Fuel Type	Emissions for 1996 Tons per year		
			Sulfur Dioxide	Carbon Dioxide	Oxides of Nitrogen
Palo Verde (2)	600	Uranium	3	3	3
Four Corners	104	Coal	1431	*	1610
Newman	482	Natural Gas	6	462	2633
Rio Grande	246	Natural Gas	3	675	1229
Copper	68	Natural Gas	0.42	117	141

- (1) Each plant may contain more than one generating unit.
- (2) Emissions to the atmosphere are nonexistent during routine operations. The impact to the environment is in the form of disposal/storage issues associated with radioactive waste.
- * The Four Corners plant is not required to track carbon dioxide data because the plant is located on an Indian reservation. As such, no information is available.

Exhibit 9

COST COMPARISON OF RESOURCE OPTIONS (ALL COSTS ARE IN 1996 \$)						
Technology Description	Typical Size (MW)	% Available During Peak	Estimated % of Time Unit Will Operate	What it Costs to Build (\$/kW)	What it Costs to Build and Operate (¢/kWh)	Time to Construct (Months)
Natural Gas Fueled Units:						
Simple Cycle Combustion Turbine	55 to 135	95 to 100	20 to 25	\$350 to \$550	6¢ to 8¢	24 to 30
Combined Cycle Combustion And Steam Turbine	110 to 255	95 to 100	75 to 80	\$600 to \$970	3¢ to 5¢	30 to 48
Solid Fuel Units:						
Plant Burning Western Coal	100 to 310	90 to 100	80 to 90	\$1,050 to \$1,350	5¢ to 7¢	54 to 72
Purchased Power:						
Buy and Transport	variable	100	NA	\$40 to \$80/yr ⁽¹⁾	2¢ to 5¢	—
Renewable Technologies:						
Wind Generation	10 to 50	< 10	25 to 40	\$900 to \$1,200 ⁽²⁾	5¢ to 10¢	12 to 36
Solar-Photovoltaic Fixed Flat Plate	1 to 50	40 to 50	25 to 40	\$2,350 to \$3,300	13¢ to 19¢	24 to 36
Increase Efficiency:						
Note: This includes options such as residential appliances, heating and cooling equipment, commercial lighting, heating and cooling, and industrial motor replacement and variable speed motors.	1 to 10 ⁽³⁾	100	100	\$450 to \$1,300	N/A	1 to 24

Note: These cost estimates are based on El Paso Electric's 1996 Alternative Technologies Study.

(1) This is an annual cost; the other numbers in this column are one-time investment costs.

(2) These costs include an investment tax credit of 1.2¢/kWh which is due to expire in July 1999 if units are not in operation.

(3) Efficiency improvements can be made in very small increments. However, to be considered as a resource option, a maximum of 1 MW is assumed to be cost effective.

Traditional Generation Sources and New Developments

Introduction

Currently, the two most common fuels for electric generation are natural gas and coal. Both are hydrocarbons formed by plants covered up millions of years ago and are referred to as "fossil fuels." Both are finite, meaning some day we will run out, but there is great debate about how much coal and natural gas remains.

Fossil-fuel power plants often produce more emissions than other options. The U.S. Environmental Protection Agency (EPA) and the Texas Natural Resources Conservation Commission (TNRCC) regulate the amount of emissions allowed. There are pollution-control equipment costs associated with building a new fossil fuel power plant and pollution considerations associated with buying from existing plants.

Like coal, the use of nuclear generation was extensively pursued following the 1973-74 energy crisis when natural gas prices skyrocketed and utilities were incorporating fuel diversity into their strategic plans. In 1978, the Power Plant and Industrial Fuel Use Act prohibited the use of natural gas as a boiler fuel, which led to increased use of nuclear power. But nuclear generation suffered a huge setback following the Three Mile Island nuclear incident in Pennsylvania in 1979. Although a proven technology, nuclear generation is not considered an option at this time.

Natural Gas

There are three basic types of conventional natural gas technology:

- Steam generator-steam turbine
- Simple cycle gas turbines
- Combined cycle gas turbines

Traditional generators linked with steam turbines have been the technology of choice for large natural gas plants, in the range of 50-400 megawatts. Their size and the time necessary to start up makes them less useful for start-and-stop operation in meeting peak loads. Conversely, their efficiency in fuel justifies the higher equipment costs when the unit is run continuously.

Simple cycle units are the least expensive to construct and use energy directly converted from natural gas. They can be turned off and on easily and, therefore, are ideal for use during peak times of the day or peak periods of the year. Although they use more fuel to generate one unit of electricity, it is often economical to use a simple cycle unit if it is only needed during peak times.

Combined cycle units capture the exhaust gas from the simple cycle gas turbine and use it to make steam to power a second generator. The tradeoff is the cost of equipment versus the cost of fuel. Combined cycle units are more expensive to build and maintain but are more efficient and have lower fuel costs than a simple cycle unit.

Previous federal law restricting the use of natural gas as a boiler fuel was changed to allow utilities and others to construct new power plants using natural gas and repower existing power plants fueled by natural gas. This factor, along with a surplus of inexpensive natural gas, has made gas generation the technology of choice for most recently constructed generation plants.

Below are some of the advantages and disadvantages of natural gas generation.

Advantages of Natural Gas Generation

- Least expensive to construct
- Flexibility in size
- Short construction time
- Burns cleaner than other fossil fuels
- Less pollution-control equipment required
- Fuel is piped directly to the plant; no storage or fuel handling required

Disadvantages of Natural Gas Generation

- Availability and price of natural gas can shift suddenly
- Air emissions, NO_x, CO₂, SO₂
- Some types of natural gas plants may require large amounts of water for cooling

Coal

Using coal to generate electricity is a tested and proven technology, and the fuel is relatively inexpensive. The newest coal plants use advanced fluidized bed combustion in order to reduce environmental harm. Additionally, plants can be designed to convert coal into natural gas, if necessary. Below are some of the advantages and disadvantages of coal use.

Advantages of Coal Generation

- Historically, coal has been a cheaper fuel than natural gas
- More stable fuel price

Disadvantages of Coal Generation

- Is the main fossil fuel that produces SO₂ and NO_x, which are linked to acid rain
- Is the primary source of CO₂, which may cause global warming
- Depending on size and location, may take two to three times as long to construct as a natural gas plant
- More expensive capital costs
- Greater risk that construction will cost more than anticipated
- Needs more fuel handling and pollution control equipment, including coal and coal-ash handling facilities
- Needs water for cooling and disposal facilities for ash

Distributed Generation

The use of small generating units at or near the site of electric utility customers is called "distributed generation". Distributed generation can include on-site renewable energy, such as solar panels or fuel cells. It can also include combustion-driven generation such as diesel units.

Today, distributed generation is already used at places that need to have backup power: hospitals, airports, water treatment plants and computer centers. These facilities typically have a generator available for use in case of a power failure. Customers who are billed, in part, based on their peak usage may use distributed generation to ensure that they do not set a new peak and increase their monthly bill.

Below are some advantages and disadvantages of distributed generation.

Advantages of Distributed Generation

- Less investment in transmission lines
- Reduces amount of "line losses" that increase as the distance of the transmission and distribution lines increase

Disadvantages of Distributed Generation

- Costs vary widely depending on size, location and application of the resource
- Smaller diesel units can have greater emissions and are often installed in urban areas where the impact is the greatest

New Developments

Several new developments in generation look promising but are still in an experimental phase. Most prominent in this category are fuel cells, which generate electricity with a chemical reaction similar to a battery and can use natural gas as fuel. They are still more expensive than either conventional generation or wind power, but their design and economics are rapidly improving. Their advantages are small size, efficient fuel use, low emissions, reliable output and quiet operation. Most see their first applications as generation sources for remote individual buildings and places that require high power quality.

One problem with electricity is that it is difficult to store. Energy storage technologies include:

- pumped hydro storage
- compressed air storage
- battery storage

At an off-peak time, when there is extra capacity and fuel prices are low, electricity is used to compress air, which is stored in an underground cavern, or to pump water, which is stored in an

elevated reservoir. When the peak arrives, the compressed air or water is released, which, in turn, drives a turbine to generate electricity on peak.

Off-peak power can also be used to charge batteries. Several utilities have built large, multi-megawatt energy storage demonstration projects using special batteries, but there does not appear to be much activity to date devoted to commercializing such facilities.

Existing Resources

Traditionally, integrated resource plans have focused on the development of an appropriate mixture of new resources to meet load growth. However, another important factor to consider is how to utilize existing resources. In particular, consideration should be given to the potential for efficiency and productivity improvements, repowering and refurbishment of existing power plants, and extending the life of power plants. The question of existing unit retirement is complex and should be analyzed based upon the value to the system as compared to other available alternatives.

Repowering and refurbishing an existing resource can be described as follows. EPE could add a simple cycle gas turbine to one of its existing steam generator units, effectively creating a combined-cycle unit. This refurbished unit could provide more capacity, run more efficiently with fewer pollutants, and use existing infrastructure and facilities. In addition, the operating life of the unit would be extended.

Repowering and refurbishment of an existing unit can be performed at any time during the operating life of a unit to meet the need for additional resources or to meet higher emission restrictions.

The decision on repowering an existing unit must be made in comparison with the other alternatives. This option is listed here because most repowering would mean increased use of gas by El Paso Electric.

Competitive Bidding

The PUCT requires utilities to conduct competitive bidding to ensure that the best alternative is selected. If El Paso Electric wants to build a new power plant, it must demonstrate that the proposed plant is better and less costly than any competitor's proposal.

Exhibit 10

GAS - VS - COAL EMISSION RATE COMPARISON OF EPE UNITS				
	No _x ⁽¹⁾ lb/MMBtu	CO ⁽¹⁾ lb/MMBtu	SO ₂ ⁽¹⁾ lb/MMBtu	CO ₂ ⁽²⁾ Ton/MWH
Copper Plant - Gas Fired	0.24	0.1	0.0006	5.2
Rio Grande Plant - Gas Fired				
No. 6	0.22	0.08	0.006	.66
No. 7	0.26	0.14	0.006	.64
No. 8	0.26	0.17	0.006	.62
Newman Plant - Gas Fired				
No. 1	0.3	0.02	0.0006	.67
No. 2	0.31	0.001	0.0006	.63
No. 3	0.41	0.08	0.0006	.63
No. 4	0.14	0.06	0.0006	.58
Four Corners Plant - Coal Fired				
No. 4	0.54	N/A	0.48	.97
No. 5	0.54	N/A	0.48	.97

Note:
1. Units in lb of pollutant per million BTUs of fuel fired.
2. Units in tons per megawatt-hour since CO₂ is primary component involving any fossil fuels.

Purchased Power and the Wholesale Market

Introduction

Another group of options available is to buy electricity from outside the service territory and transport the electricity into El Paso Electric's system. There is currently a plentiful supply of power available in many parts of the country, including the Southwest, although the ability to transport the power is limited by transmission capability. This option has the advantage of allowing El Paso Electric to purchase only what it needs to meet growth on a year-to-year basis and would not necessarily result in a long-term investment that could become uneconomical in the future. El Paso Electric can delay purchasing power until it is needed and when El Paso Electric can better predict its own needs. This provides El Paso Electric more flexibility as opposed to committing to build a new generating unit many years before it is needed.

Expanding Wholesale Market

The wholesale market includes sales to electric utilities for resale to their customers. The electric utility industry is changing rapidly, with more suppliers entering the electricity market. These suppliers include power marketers, independent power producers, other utilities and cogenerators. As these suppliers enter the market, there may be opportunities to buy new sources of power at overall lower market prices.

Many experts believe that the wholesale electricity market will continue to develop as more utilities rely on this option to meet the growing needs of their customers. Some suppliers have already begun to construct new generation facilities with the expectation that they will be able to sell power into this market in the future.

Purchasing power through the wholesale market has the advantage of allowing El Paso Electric to purchase only what it needs to meet growth on a year-to-year basis. This avoids investment in more long-term facilities, which would be expected to be used and paid for over a 30-year period.

A disadvantage of purchased power is the difference in the ability to control prices on either long-term or short-term power purchases. Short-term purchases may be favorably priced right now because there is a surplus. However, over the long term, the price and supply of this power will be more uncertain as it will depend on market forces such as supply and demand. Another disadvantage is the reliability of the purchased power may be compromised unless backup arrangements are made with the entity delivering the power.

Who Is Selling The Power?

In addition to other utilities, El Paso Electric can turn to several non-utility sources to purchase wholesale electricity.

Power Marketers

A new type of merchant has recently emerged in the electricity business. Just like the gas marketers that currently sell most of the natural gas traded at the wholesale level, power marketers buy and sell bulk electricity, generally without owning generation facilities themselves. For example, power marketers can buy small amounts of surplus electricity from a number of sources and put them into a package to resell to others. A few years ago, there were only a handful of registered power marketers in the U.S. Today, there are over 200 registered power marketers.

Power marketers usually have a wide portfolio of electricity contracts available to sell. They are often able to sell a package that exactly meets the purchaser's requirements at a competitive price. The disadvantages are that the price and availability will be subject to market forces such as supply and demand in the long term, plus the potential to compromise reliability of the power being provided.

Independent Power Producers (IPPs)

IPPs are businesses that own generation facilities for the production and sale of electricity to wholesale customers such as electric utilities. There are several hundred large independent power producers in the U.S. today. Some examples of IPPs are industrial plants, subsidiaries of construction

companies, and utility companies competing outside their service territory. Electric generating facilities owned by IPPs are not limited to gas and coal generation. IPPs may also build generating units powered by renewables such as wind or solar energy. IPP generating facilities could be built in the El Paso area, providing an economic benefit to the area.

Cogeneration

Another source of electricity available to El Paso Electric is from cogeneration plants. These are plants that produce steam and use it both to generate electricity and in manufacturing processes. Usually, these plants exist at major industrial sites, refineries or chemical plants.

The primary advantage of cogeneration is energy efficiency, because more of the heat content of the steam is captured for energy use. Another advantage is economic, since cogeneration can help make the industrial steam user more competitive. The electricity produced by the cogeneration facility could be purchased by El Paso Electric under a short- or long-term purchased power contract.

The main disadvantage of cogeneration is that purchases could be interrupted or curtailed if the steam host closes its manufacturing operation, moves or goes out of business. Thus, El Paso Electric's reliability could be affected.

Transmission Capacity And Delivery

A combination of transmission and distribution lines carries electricity to the consumer. First, transmission lines transport electricity long distances and usually carry power to a substation. Then, the distribution system takes over and carries the electricity from the substation to the customer. All power plants require transmission lines to deliver electricity out of the plant and onto the distribution network. The transmission system can be divided into two parts, the regional system and the local system.

The regional transmission system operates at 345 kv and brings power long distances from Palo Verde in Arizona or Four Corners in New Mexico. It also carries the power purchased from outside El Paso Electric's system. If the resource mix calls for large amounts of purchased power, then the regional transmission system may require upgrades. Upgrades can mean new lines or reinforced lines. Upgrades can also include new technologies such as superconducting magnetic storage that improves power flow. The local system operates at lower voltages (69 kv or 115 kv) and delivers power to the 76 El Paso Electric substations. Distribution lines take over from there to bring the power to your home or business.

Much like the decision to repower old generation plants, building or upgrading new transmission lines should be weighed against the alternatives. At times, transmission line construction can be controversial if landowners or affected citizens object to the lines. Their concerns must also be taken into consideration. As a consequence, this factors into the overall pricing structure of this alternative.

Competitive Bidding

As with power plant construction, the PUCT requires that suppliers bid against one another. In this manner, El Paso Electric's long-term needs can be satisfied by parties who have demonstrated that their proposal is best.

Efficiency Programs

It is feasible and often less expensive to meet some of the need for power by using the energy we already have more efficiently. The purpose of energy efficiency programs is to decrease the amount of energy used without compromising the quality of service a customer receives. Energy efficiency can mean helping homes and businesses use less electricity with insulation, window shading, energy-efficient lighting, energy-efficient refrigerators, water heaters and using more solar energy in buildings.

These energy efficiency investments take place on the customer's side of the meter where the demand is, and, therefore, are known as "demand-side" resources. The use of demand-side management (DSM) to meet new generation needs has boomed in the last 10 years as new technology in lighting, motor drives and controls has entered the market. Some DSM options, such as "peak clipping" and pricing options, may require a change in lifestyle or business practice. Some experts believe as much as 40% of energy use could be met with use of efficiency programs.

Energy Efficiency

Most customers can conserve electricity and still receive the same level of service. This is accomplished by making building structures more efficient by installing insulation, caulking and weather-stripping, solar screens, roof overhangs and other measures that reduce cooling and heating needs. Another way to reduce electricity use is to install the most efficient equipment available for cooling and heating. Equally important are water heaters, refrigerators, lighting fixtures and other equipment. One example of an energy efficiency program is the El Paso Electric E-Lite program that provides advice on high-efficiency lighting technologies. Improving building structure reduces the size of cooling and heating equipment needed. This reduces peak demand and lowers customer bills. The utility and its customers benefit by delaying the need for new power plants. Emissions are reduced because less fuel is burned. Fuel costs decrease by running (using) existing power plants less frequently.

Most customers have a great deal of control over their energy consumption. They can exercise control in choosing appliances and should consider energy efficiency ratings. In a survey conducted by El Paso Electric, only 58% of customers stated that they consider efficiency ratings when purchasing an appliance. In El Paso, customers can decide whether to have evaporative cooling or refrigerated air. They can select among natural gas, electricity, and solar energy for heating and water heating. Data on appliance saturation surveys shows that more than 76% of El Paso Electric customers stated they have evaporative cooling. Taken together, these decisions can have a significant impact on electric bills.

Terms

Demand-side - make improvements on the customer's side of the meter, such as insulation or solar screens to reduce energy use.

Supply-side - add resources on the utility's side of the meter, such as a new purchased power contract to meet generating capacity needs.

Pricing Options

Pricing can also influence when and how customers consume electricity. Higher prices during the summer often influence the amount of electricity used during that time of the year. These higher prices are necessary, since it costs more to meet the demand in the summer when the peak is higher. There is also "time-of-use" pricing available for some classes of customers that reflects the changing cost of energy production through the hours of the day and the days of the week. These changes in price can also influence the on-peak usage by customers.

There are other pricing options that can help reduce peak consumption and demand:

- Interruptible and curtailable rates are offered to large industrial customers where the customers accept the increased probability of interruption or curtailment of electric service in return for a lower rate.
- Real Time Pricing is another alternative that could be implemented in the near future for different

customer classes that have the ability to adjust operations on a daily or hourly basis to move load from the higher cost hours to the lower cost hours of energy production. This can lower their electrical bill and provide load-leveling benefits to the Company.

Historically, these programs have been aimed at large commercial and industrial classes.

Peak Clipping

Another option to meet the need for power is to focus on the peak usage times—those hours and days when total usage is at its highest level—and try to reduce demand at these times when electric costs are highest. Normally, during the weekdays of June, July and August during the hours from noon to five in the evening, power usage is more likely to be at its highest for the year.

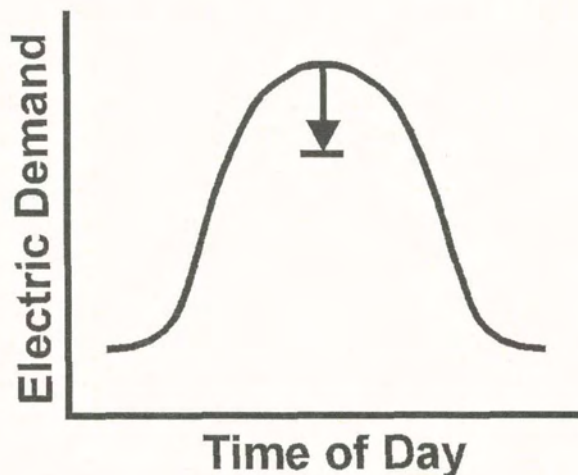
Some programs reduce peak load by charging higher rates during periods of high demand. Others reward customers for allowing the utility to interrupt their service when necessary. One example of a peak-clipping program is a control system that sets air conditioners, water heaters and refrigerators to run at different times. The customer seldom notices, but peak load of the overall system is reduced.

Programs must be carefully designed, because, often, residential and commercial users may have difficulty investing in equipment with high up-front costs, getting essential information, or taking advantage of programs aimed at peak reduction because it would mean changing their lifestyle.

Industrial customers are often in a better position to make use of these programs because they can either switch to backup generation that they own themselves or adjust their work processes to reduce demand. Some people say that focusing only on reducing peak usage misses a number of other cost-effective conservation measures that reduce utility costs, conserve fuel and are beneficial to the environment.

As electricity use in the region increases, El Paso Electric's ability to meet this peak demand will diminish. Eventually, the Company will be unable to meet the need without more resources. Since peak usage is usually what causes utilities to need more capacity, reducing peak loads can delay the need.

Peak Clipping



Load Leveling

Some energy management programs are designed to even out the levels at which customers use electricity. Load leveling is also referred to as "load shifting" or "valley filling".

An example of load shifting is the El Paso Electric Thermal Energy Storage program (TES). The program encourages owners of commercial facilities to install a thermal energy storage system. The system chills water or creates ice overnight, which is then used during the day to supply air conditioning. Storage systems thereby shift the peak energy use of large buildings to off-peak hours.

Another type of program used to valley fill on the power system is designed primarily to increase off-season sales. Sales are generally lower in the winter, when the power plants built to meet peaks in summer produce less power to meet reduced demand. An increase in units of electricity sold

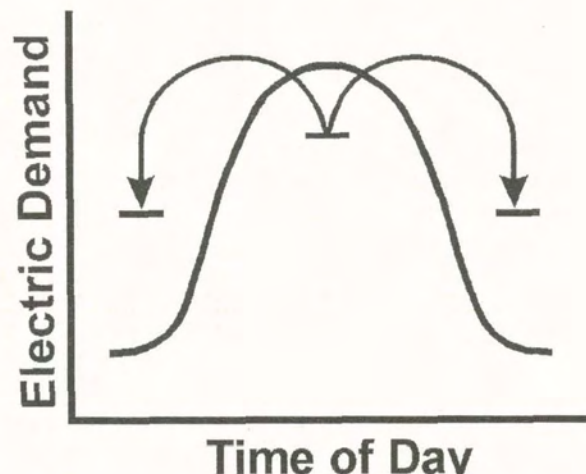
without the need for new generating plants means the rate per unit can go down.

Most utility companies favor load leveling. Load leveling results in more regular power plant operating hours and sometimes increases the kilowatt-hours sold from these plants. Consumer and environmental groups, however, are less enthusiastic about load leveling. They reason that load leveling programs increase electricity use, which can mean higher bills for customers even if the rate per kilowatt-hour may be lower.

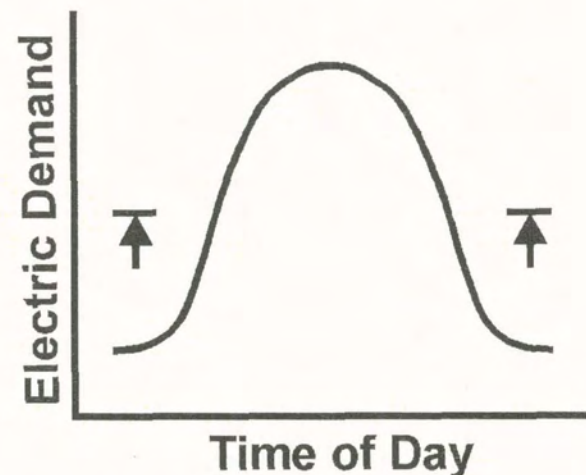
Load leveling can be accomplished by waiting several hours to use energy. Some examples include washing clothes at night and starting an evening shift at an industrial plant.

Again, by lowering the peak, these programs make it possible to avoid the addition of generation. Another advantage of these programs is that customers can still use power for all the things they are accustomed to doing, but they just do so at different times. Some critics say that load leveling programs do not reduce fuel consumption and, therefore, do not help the environment. Proponents, on the other hand, say that such programs allow the utility to use its resources more efficiently and that overall electricity bills remain the same or are reduced.

Load Shifting



Valley Filling



Existing EPE Demand-Side Programs

Residential

Residential DSM programs serve residential and low-income customers. The Residential Audit Program provides a free on-site energy audit by El Paso Electric energy auditors. Free installation of weatherization and energy efficiency items is performed by an outside contractor. The materials include caulk, door sweeps, door weather-stripping, electric outlet gaskets, water flow restrictors, and refrigerator coil cleaning brushes. Since its introduction in 1989, the energy audit program has helped 2,500 families become more energy efficient. A total of 1,200 audits are scheduled this year, including 600 for low-income customers. El Paso Electric has had success with a program called "Block Party", where auditors knock on doors and concentrate in specific neighborhoods. The Low Income eEfficiency Tools (LIFT) program provides similar services to qualified low-income customers. More than 600 families have been served by this program since December 1994.

Commercial And Industrial

The Commercial and Industrial Energy Services Program (CIES) is an umbrella program whereby El Paso Electric provides energy analyses and recommendations to commercial, industrial and municipal customers. Energy auditors apply specific programs within CIES, based on customer needs, that enable customers to achieve their energy goals and objectives. Since 1988, 14.7 megawatts have been saved through these services.

The E-Lite Program provides recommendations on high efficiency lighting technologies. The pilot program Customized Efficiency Options (CEO) provides recommendations on energy efficiency and load management technologies for heating, ventilation and air conditioning (HVAC), motors, and industrial processes. The Thermal Energy Storage (TES) Program encourages owners of commercial facilities to install TES systems, which produce chilled water, ice or other cool storage media such as salts during off-peak periods. The media are then used to provide cooling during periods of peak demand in place of additional or larger chillers. This process shifts consumption from peak to off-peak periods.

Competitive Bidding

The PUCT has expressed a preference for competitive bidding as a way to assure demand-side and efficiency programs have the lowest guaranteed cost. This means that, if the decision were made to meet a portion of the new generation requirement through energy efficiency, then El Paso Electric would publish a request for bids to supply the efficiency programs. According to PUCT rules, the Company can participate in the bidding process if it chooses, but only as one of the bidders, and then only with the participation of an outside evaluator.

Fair Treatment

Fair treatment for different customer types has long been an issue with demand-side programs. Many of the programs require some action on the part of the consumer, such as an investment in more efficient equipment or appliances. The Company, in turn, pays a portion of that investment as an incentive, and all customers help to pay the Company's costs in their rates. Customer groups with low or fixed incomes often lack the funds to participate in such programs even though it might be in their long-term economic benefit.

Another group that can easily be left out of these programs is tenants, who represent about 42 percent of El Paso Electric's residential customers. Investments, such as increased insulation that might make sense to the owner of a structure, may make little sense to a person who plans to spend a year or less living in an apartment.

Currently, the Company budgets \$1,200,000 for DSM programs and allocates 5% of the budget for its low-income DSM program, namely LIFT. In addition, Company funds for LIFT are matched by state funds in a partnership with UTEP under a one-year joint proposal. One third of the El Paso County population have incomes below federal poverty guidelines.

Under current regulations, El Paso Electric must design its DSM programs so that low-income customers and tenants benefit. One of the issues in the Deliberative Polling™ exercise is the extent to which customers are willing to support programs designed to achieve fair treatment of all customer groups.

Advocates of low-income DSM programs state that electricity is essential to life and that DSM programs for those on low or fixed incomes help make utility services more affordable. They state that, while it is true that some low-income programs may cost more per unit, there are ways to design the programs to minimize or eliminate the additional costs. For example, utilities can coordinate the delivery of low-income programs with other government-funded providers to eliminate duplication and provide comprehensive energy assistance programs.

Glossary	
Increase Efficiency and Conservation	Reducing electricity use without changing lifestyle.
Peak Reduction	Using less electricity during peak hours. Usually this term implies a change in lifestyle or business practices.
Thermal Energy Storage (TES)	Changes electricity into another energy form such as chilled or heated water for use in air conditioning or heating at a later time.
Interruptible Rate	A rate structure where customers accept increased probability of interruption of electric service in return for lower rates.
Base Load	The level of utility load expected to be present most of the time. A base load plant is designed for continuous operation and generates electricity at least 60 percent of the time.
Intermediate Load	The portion of utility load that varies between base and peak load. Intermediate plants typically operate 20 to 60 percent of the time.
Peak Load	A utility's maximum load during hours when rate of usage is the highest. Peaking plants typically operate less than 20 percent of the time.

TEXAS CURRENT RATES							
	Customer Charge	Demand Charge (\$/kW)	Energy Charge (\$/kWh)	Fuel Charge (\$/kWh)	Average Monthly Usage		Average Monthly Bill (\$)
					kW	kWh	
Rate 01 - Residential	\$ 4.50			\$ 0.01388		495	
Summer (Jun - Sept)			\$ 0.08858				\$ 55.22
Winter (Oct - May)			\$ 0.08358				\$ 52.74
Rate 02 - Small Commercial	\$ 5.50			\$ 0.01388		1,024	\$ 142.68
First Block (1st 1,500 kWh)			\$ 0.12008				
Second Block (All Other kWh)			\$ 0.10361				
Rate 11 - Municipal Pumping	\$23.25		\$ 0.05383	\$ 0.01388		46,870	\$ 3,196.82
Rate 24 - General Service	\$13.00	\$ 12.75					
First Block (1st 200hrs * kW)			\$ 0.04878				
Second Block (Next 150hrs * kW)			\$ 0.03819				
Third Block (All Other kWh)			\$ 0.02440				
Secondary (480 volts or less)				\$ 0.01388	68	24,379	\$ 2,285.45
Primary (2,400 volts or greater)				\$ 0.01312	68	24,379	\$ 2,266.93
Rate 25 - Large Power	\$100.00	\$ 20.25					
First Block (1st 400hrs * kW)			\$ 0.02320				
Second Block (All other kWh)			\$ 0.00615				
Secondary (480 volts or less)				\$ 0.01388	882	396,809	\$ 31,923.82
Primary (2,400 volts or greater)				\$ 0.01312	882	396,809	\$ 31,622.25
Rate 29 - Transmission Voltage		\$ 21.25	\$ 0.00645	\$ 0.01287	22,774	13,964,000	\$ 753,731.98
Rate 41 - City and County	\$15.25	\$ 16.75					
First Block (1st 3,000 kWh)			\$ 0.12209				
Second Block (All Other kWh)			\$ 0.01293				
Secondary (480 volts or less)				\$ 0.01388	100	15,134	\$ 2,423.47
Primary (2,400 volts or greater)				\$ 0.01312	100	15,134	\$ 2,411.97

Renewable Energy

Introduction

The idea behind renewable energy is the difference between burning fuel that is finite in supply compared to using an energy source that regenerates itself or is replenishable. Obviously, if the costs were the same, utilities would use the resource that is not limited. But, costs are not the same. The cost to build facilities to take advantage of renewables such as wind power or solar power are still more expensive. This upfront cost is not usually offset by the absence of fuel costs. However, the price of renewable energy is coming down.

Proponents of renewable energy point out other factors to consider, such as environmental costs and health costs. Those who favor lowest cost energy sources have a different view. This is an area where we need your judgment.

Wind Generation

Wind turbines are beginning to come of age. In the last five years, wind power has made major technological and economic strides. It compares favorably to natural gas because its costs are more certain than natural gas. Studies show that wind energy is available the most during El Paso Electric's off-peak season or winter months. Although wind generation could be located in the El Paso region, the most favorable sites for wind generation in the area are farther east in Texas (see **Exhibit 11**).

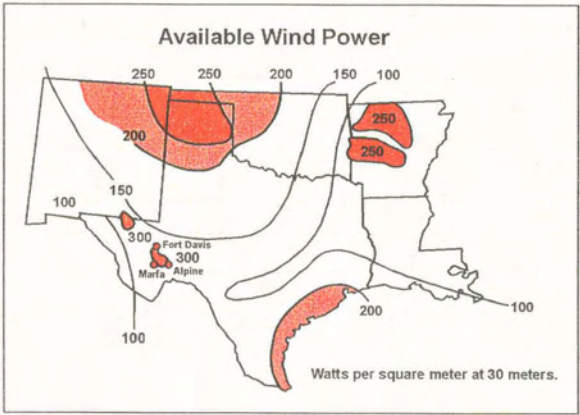
Advantages of Wind Power

- Least expensive form of renewable generation
- Flexibility in size
- Serves as a hedge against sudden shifts in the price of fossil fuels or changes in environmental regulation
- May be competitive with natural gas generation over the 30-year life cycle of the wind generation equipment
- Has no emissions
- Land use is compatible with cattle ranching or farming

Disadvantages of Wind Power

- The wind may not be blowing when power is needed, especially during peak demand periods of the summer
- Increased bird mortality due to birds flying into the blades
- Visual aesthetics
- Initial cost per kilowatt-hour is higher than fossil fuel
- Depending on the location, new transmission lines may be needed
- Requires leasing large tracts of land
- Accessibility to wind generation sites

Exhibit 11



Solar Energy

Solar power is the use of the sun's rays to meet energy needs. It can be used in many different ways to meet customer needs for heating and lighting buildings; for hot water in homes, businesses, and industry; and to make electricity, either for sale to customers or directly for use by customers.

Passive Solar and Daylighting

Passive solar building design means using a building's orientation to the sun, as well as strategic placement of windows, overhangs, and other building features, to minimize heating and cooling loads. Daylighting means designing a building in order to bring the sun's light into building spaces. This could include new reflective technologies. In both cases, electric

loads for heating, cooling and lighting can be minimized or avoided. Both of these technologies are well suited to the El Paso and Las Cruces areas.

Solar Thermal

The sun's heat can be collected to heat water or air for use in homes, businesses or industry. Where these solar collector technologies are used, they may replace or reduce the size of furnaces or water heaters or they may be used to preheat water for commercial or industrial uses. Solar water heating was popular during the 1970s and early 1980s when natural gas prices were high. Adding solar collectors for heating buildings and water may reduce the amount of electric energy required.

Solar heat can be collected on a larger scale and used to power electric generators. In certain California desert areas, large fields of reflective troughs aimed at the sun heat oil within a network of pipes. The heated oil, in turn, transfers the "solar" heat to steam generators which are used to generate electricity. These generators also use some natural gas to make electricity when the sun is not shining.

A different, smaller solar power plant is now being developed that involves mirrors focused on a special motor that uses solar heat with natural gas to produce electricity.

These solar heat electric generation technologies are in various stages of development. Some are producing experimental power for utilities today, while others need more time and testing, as well as further cost reductions before they will be ready for use by utilities.

Solar Pond

Solar ponds capture the heat of the sun's rays using layered water at different saline densities. Generally, there are three main layers in the pond. The top layer is cold and has very little salt content. The bottom layer is hot, up to 100° C (212° F), and is very salty. Separating these two layers is the important gradient zone. Here, salt content increases with depth. The stable gradient zone suppresses convection and acts as a transparent insulator, permitting sunlight to be trapped in the hot bottom layer from which useful heat may be withdrawn or stored for later use.

Solar pond development has been limited in the U.S. Currently there is only one solar pond in the U.S. capable of producing electricity. This pond is located in El Paso Electric's service territory and is operated by the University of Texas at El Paso (UTEP). It started generating electricity in September 1986, supplying the Bruce Foods Company with power and steam for its food processing operation. Additionally, the pond is used to produce fresh water from contaminated underground water. Research from the pond has been concentrated in this area rather than in the production of electricity. The greatest potential market for solar ponds in this country could be in residential space heating and industrial process heat sectors. Therefore, although there may be other applications for using solar pond energy, the technology is currently too expensive to be considered as a viable generation resource at this time.

Solar Power

Several very different technologies have the ability to transform solar energy into electricity.

Solar electric power can be created by semiconductor materials in solar panels that change sunlight into electric current. Solar panels have been used in the space program from its beginning. Photovoltaic (PV) materials have many uses in consumer products, such as calculators, lights, gate openers, and battery chargers. Other currently cost-effective uses for photovoltaic electric power include water pumping, electric power for remote homes and for any small electric load that is remote from, or hard-to-serve, by conventional electric lines.

Solar panels on rooftops can help meet a building's peak electric load, which typically occurs

when air conditioning needs are the highest and the sun is most intense.

Advantages of Solar Power

- Pollution free
- Flexibility in size
- Uses the sun's energy as a source of heat to generate electricity
- Avoids the costs and risks of fossil fuels
- Cost-effective for specialized uses

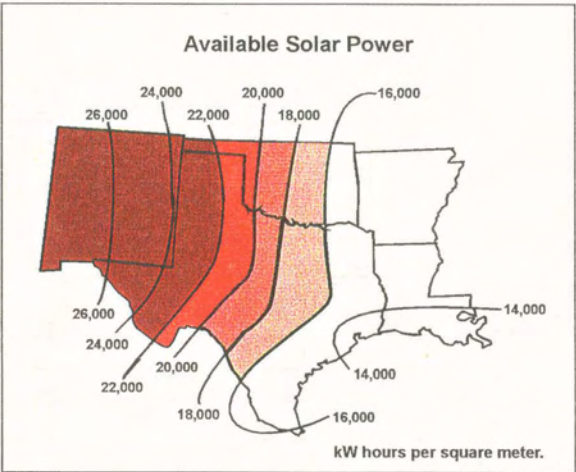
Disadvantages of Solar Power

- Capital costs are higher
- Because weather conditions may limit the amount of sunshine, the utility must combine solar energy with other types of power
- Visual aesthetics
- May require large tracts of land

Non-grid Connected Solar Applications

One of the most cost-effective solar applications is to use a solar installation in place of a more expensive line extension. The PUCT requires utilities to distribute information concerning on-site renewable energy technology alternatives to customers considering line extensions. The PUCT also maintains a list of companies that provide renewable systems and services.

Exhibit 12



Other Types Of Renewables

Biomass - In some places, agricultural products are grown and then burned to produce energy. Rice hulls, sugar cane and wood waste are examples of using waste by-products. Some species of fast growing trees are examples of a product grown specifically to produce energy. Animal manure can also be burned.

Landfill Gas - Landfills produce gas as their contents decay. Federal law requires landfills to collect and dispose of this gas (methane), which can be burned to produce electricity. Landfill gas projects tend to come in 1.5 - 3 MW sizes so they are easy to integrate into a utility system. They are generally located near an urban area because of the location of the landfill. Both technologies have advanced in the last five years. Many landfills can produce gas for 20-25 years.

Hydro Power - Hydro power is an old form of renewable energy. El Paso Electric has benefited in the past from hydro power brought in from other areas of the country.

Customer Choice Programs

Several national polls, as well as some in Texas, report that customers are willing to pay more for improved environmental performance. Customer choice programs, sometimes called "green marketing", are an attempt to allow those customers to exercise that choice. This is similar to the choice of buying recycled products.

Many electric utilities are examining the feasibility of designing green marketing programs that may be introduced in the future. The most important variable in this effort is designing a product that is attractive to customers.

Customer Choice and Change in Electric Services

In the past, the public policy dictated that electric utilities should be monopolies, with only one company given the right to serve in each area. In return for this grant of monopoly, the electric utility had to agree to build generation and lines to serve any customer who requested electric service. Rates were set by regulators on an average, all-rolled-in cost basis. As a result of changes in federal and state law, the trend is toward competition. There may be several providers of electricity, although still only one set of lines. Prices in the future may be set by competition in the marketplace. Customer choice would be a fundamental change in the electric utility industry and the prospect of change raises a number of questions, issues and opportunities. Last year, bills were introduced in both the Texas and New Mexico legislatures to allow retail competition. As a result of disagreements among various parties, neither law passed.

As part of this Deliberative Poll™, El Paso Electric wants to discuss with you some potential new services and wants to get your reaction. It also wants to discuss your impressions and suggestions for existing services to make those services more valuable to customers.

Customer Choice Issues

Equity: Will residential and small commercial customers benefit from lower prices and more service choices along with large industrial customers?

Universal Service: Will there be an affordable, basic package of noninterruptible service available to all residential customers regardless of income or location in rural areas?

Market Power: Is it necessary for existing utilities to be divested into two or three smaller companies to prevent them from controlling market prices?

Stranded Costs: If some generation plants are not competitive because the power they produce is more expensive than the market price, how will the investment for those plants be recovered; who will pay?

Reliability and Safety: Will the quality of service offered today be maintained or improved?

Consumer Protection: How will residential consumers be protected from abusive sales tactics, unfair pricing and billing practices, inappropriate disconnections and disconnections that endanger life and safety?

Environmental Protection: Will the oldest, most polluting power plants reap the highest profits, and how will energy efficiency programs and renewable energy resources be used to diversify the resource base?

Timing: When should retail competition begin?

The above are the major consumer issues in the debate. Everyone agrees on one point -- choice will affect customers. Supporters of competition say that prices will go down and services will get better. Detractors say the opposite. Many consumer groups have taken the position that they cannot support a change in the system until laws are passed to fully protect consumers. It is fair to say that how much customers benefit under any system depends on how the market is structured.

Why Is Change Taking Place?

One of the major factors driving new laws and regulations is advances in technology. Put simply, new generation costs less than old generation. Previously, it took 10K or 11K Btus of fossil-fuel energy to produce a kilowatt-hour. New generation can produce a kilowatt-hour for about 7,500 Btus. The increased efficiency results in less fuel used, less emissions and lower costs. New generation comes in smaller sizes and, therefore, is more flexible. In addition, the technology for generating electricity has been packaged so that companies other than electric utilities have the ability to operate electric generators.

Another major factor driving change is governmental directive through both law and regulation. The prevailing current economic theory on the electric utility industry is that competition yields more efficient results than does regulation. Experience with other industries indicates that granting a monopoly to a single company is a good

way to get basic facilities and investments made. But now that the basic infrastructure and facilities are in place, competition with multiple providers has the potential for producing more efficient day-to-day results.

Status In Texas

In 1995, the Texas Legislature made the wholesale market for electricity in Texas competitive. In other words, as a result of a change in the law, any producer of electricity can sell electricity in Texas for resale. Under the current statute, retail sales to end users can only be made by regulated utilities.

The PUCT believes that electric markets should be more competitive and has taken several actions within its power to stimulate competition. For example, the PUCT passed rules requiring utilities to transport power in Texas for competitive suppliers. A power supplier no longer has to own transmission lines to deliver power to a wholesale customer in Texas. The PUCT wants to protect customers from funding competitive and profitable services that utilities provide. Their preference is for affiliates to provide these services.

In the 1997 Texas legislative session that ended in June, movement toward competition in the electric industry was a major topic. A bill emerged that would have required competition in the year 2001. The bill did not pass, but it may be an indication of time frames of competition to come.

Status In New Mexico

In New Mexico, legislation was introduced to allow for competition, but it also failed. In late 1995, the NMPUC undertook an investigation of electric industry restructuring. The NMPUC has collected comments, convened work sessions during the last half of 1996, and recently initiated a collaborative process in an attempt to arrive at a consensus around which restructuring legislation can be drafted. The collaborative process involves participants representing investor-owned utilities like El Paso Electric, rural electric cooperatives, municipal electric utilities, various residential and small commercial consumer groups and advocates, industrial consumers of electricity, low-income advocates, environmental advocates and the NMPUC Staff. While complete restructuring of the industry includes numerous issues, the collaborative participants agreed to concentrate on the most difficult issues of stranded costs and market power. To date, these two issues are still major hurdles, and, for some collaborative participants, virtually insurmountable hurdles. The collaborative process has until September 15, 1997 to determine whether there will be significant consensus to form the basis of a legislative proposal for the 1998 legislative session.

Position Of El Paso Electric

El Paso Electric Company, like all electric utilities in the U.S., is a monopoly provider with strict requirements to provide service to all who request it. As such, the Company is caught in the middle of a world moving towards competition. As noted earlier, the 10-year rate freeze, the plan by which the Company emerged from bankruptcy, further complicates the situation.

El Paso Electric is willing to support competition. In return, the Company seeks the following assurances:

- Those companies who wish to compete must share the same obligations as El Paso Electric.
- The investors who put up the money to build current facilities must have a reasonable chance to recover their money.
- Those programs and services required because they are good for society, but that might not be demanded by the marketplace, should be paid for by everyone.

Position Of Other Parties

As you would expect, parties to the discussion about competition in the electric utility industry have different positions based upon their interests. Potential competitors seek the opportunity to compete. Large customers see competition as a way to lower costs. Environmental and social advocates are concerned about costs that are not represented in market prices.

By far, the issue that has generated the most debate and the issue that has been given the most coverage in the press is the issue of stranded costs. The PUCT issued a staff report in October of 1996 estimating that stranded costs in Texas could range from a (negative) (\$2.9) billion to a high of \$22 billion but will most likely be from \$3.8 billion to \$14 billion. Estimates for El Paso Electric's Texas jurisdictional stranded costs range from \$401 million to \$1.3 billion. The position of consumer and environmental representatives has been that the PUCT study presents some interesting information but it would be inappropriate to base any cost recovery plan on those estimates. They believe any discussion of payment of stranded costs should begin after competition is in effect for two or three years so that discussions can be based on actual rather than estimated costs.

The utility companies, on the other hand, believe strongly in the right and practical need to recover their stranded costs before competition begins and the current market structure changes irreparably. Competitors state that allowing utilities to recover their stranded costs gives them an unfair advantage in the market and that true competition cannot occur if stranded costs are fully recovered. Industrial customers have been the main proponents of competition to lower rates. For the most part, they are unwilling to agree to pay any charges that will increase their rates under competition. Consumers who opposed the construction of nuclear power plants (the source of most stranded cost) say that stockholders, not consumers, should pay any stranded costs created by cost overruns and nuclear plant management. Environmentalists are worried that paying the stranded debt on old, inefficient and polluting plants which are not subject to new emission standards will make them the cheapest and most profitable plants to operate in a competitive system.

How Might Customers Be Affected

Large customers will most likely be the first to see their choices expand under competition. They are the least expensive to serve. Middle-sized customers, such as stores and office buildings, will probably be the next to see competitive choices. Small customers, such as residences, will see competitive choices but probably as part of a group. A neighborhood or subdivision might aggregate its residents into a purchaser able to exercise competitive power.

Those who have concerns about competition point out that competition is neither orderly nor kind. Some competitors prosper; some competitors fail. Customers who were accustomed to protection by the regulatory agency may have to protect themselves in a competitive environment. The flip side to having more competitive choices is that customers must take the time to educate themselves in order to exercise reasoned judgment. Those who see benefits in telephone competition, airline competition, or trucking competition are likely to favor electric competition. Those who long for comforts such as a unified telephone system may have some concerns.

Consumers say that the long-distance telephone industry is the best example of why standards are needed in the electric industry to protect consumers. They believe prices should be stated in a uniform manner so they can be compared. Consumers argue that all suppliers under competition should be bound by the same billing and contracting standards and that the PUCT should be required to provide extensive educational information to the public on changes in the industry.

In the future, most expect the generation of electricity to be competitive. The transmission and distribution of electricity to the home or business is likely to remain a monopoly. The provision of billing services, meter reading and other customer services are likely to be competitive with multiple suppliers.

Possible New Competitive Services

The following descriptions are for services that might be provided in a competitive environment. They are intended to give participants a feel for the range of possible services and are certainly not a complete list. One of the advantages of competition is that it stimulates innovation. It is likely that some of the most successful competitive services have still not been developed. The purpose of introducing this list is to get some early feedback from customers on directions and services they might like to see.

Energy Efficient Mortgages

With this service, the provider would loan a higher percentage of the purchase price to someone buying a home that has been certified as energy efficient or who is making a major energy efficient renovation to their home. Because the home buyer is borrowing more money, the mortgage payments would probably be higher. However, the monthly energy bills would be lower than average, so the monthly overhead would be equal to, or less than, that of a conventional loan.

Mail Order Catalog

A possible service includes offering energy efficiency and other home comfort products through a mail order catalog. Such items could include compact fluorescent light bulbs, energy efficient light fixtures, set-back thermostats, and water conservation products. The catalog could assist customers in making purchasing decisions by briefly explaining how the products can help reduce energy bills. Customers might be able to pay for these items in installments through their electric bills.

Interruptible Rates

Current energy rate structures allow customers to take as much electricity as they want when they want it. The option remains with the customer, and the company retains the obligation to serve and, therefore, maintains a reserve margin for unexpected use. Reserves, like excess inventory, are expensive. A cheaper way to serve customers is to limit their ability to use electricity at the peak (or most expensive) times. The most common form of this concept is an interruptible rate. Under interruptible rates, the utility company offers the interruptible customer electricity at a lower price with the right to interrupt the customer at those times when the system is nearing its capacity. In essence, this is a lower quality service. The customer, at its option, receives a beneficial rate for giving up its right to unlimited service. In a competitive environment, there will likely be a broader range of service quality levels. Lesser levels of service quality would theoretically offer cheaper prices. Competitors could match customers with differing load profiles to limit their risk. Pricing structures and service quality levels would be more complex than they are currently, but would also offer customers a broader range of choices.

Appliance Warranties

One competitive program that has already been seen across the country is appliance warranties and appliance services provided by the energy supplier. This is an example of energy suppliers expanding into the uses made of the energy they supply. Under this program, the utility or a competitive energy supplier also provides warranties for major home appliances. The cost of the warranty depends on which appliances are covered. The advantage to the customer is single supplier convenience. Customers already have a relationship with the energy supplier, and the energy supplier already has trucks and crews in the field.

Real Time Pricing And Metering And Control Options

Although existing rates average costs over time, the cost to supply energy varies at different times of the day. Energy supplied at periods of low demand is the cheapest, and energy supplied at periods of high demand is the most expensive. There are also variations throughout the year. One possible competitive service is to bill customers at their option for the actual cost at the time they choose to consume energy. This option is known as "real time pricing." For customers to have sufficient information to make informed decisions, this program would require more advanced metering. Programs like this are already available for large customers in some parts of the country but are also possible for small customers based upon new in-home metering technologies.

The metering and control options are similar to real time pricing. Current meters are based on technology that is 30-50 years old. New metering technology would allow customers greater information and control over their bills. In some of the systems demonstrated to date, customers can program specific appliances to turn on and off in relation to favorable energy prices.

Designating A Fuel Type

Some customers have a preference for a particular energy source. The most common example is those customers who believe more energy should be provided from renewable resources or sources more beneficial to the environment. Under this type of product offering, customers might designate all or a portion of their energy to be supplied from renewable sources. For example, in Colorado, customers can purchase wind power in blocks of 100 kWh for an additional \$2.50 over the cost of coal-generated power.

Who Are The Competitors?

The participants in the competitive environment will likely be a combination of existing electric companies operating outside their current service boundaries and new companies that move into the energy services market. Some will offer a full range of energy services while others will concentrate in specific market niches. Currently, many of the companies poised for competition are non-regulated arms of utility companies. In addition, there is a group of companies expanding from related industries such as the natural gas industry. Companies that will concentrate in market segments, such as generation, include large industrial companies already generating electricity. Many of the companies who will collect small amounts of power and market them as a package come out of either the natural gas industry or the financial services industry. In this case, marketing is often similar to selling stocks and commodity futures.

Summing Up: This is Where We Need Your Judgment

In Planning For The Lowest Cost For New Resources - What Do You Think Is Important?

According to state law, the goal of the planning process is to produce electricity at the lowest reasonable system cost without sacrificing reliability. The concept of lowest reasonable cost seems straightforward. However, as you know from any major purchasing decision, there are many ways to think about cost.

Let's use the automobile purchase as an example. There is the upfront cost of the automobile and the monthly cost to maintain and operate the car. If you drive long distances, gas mileage is a real concern. There is the decision whether to purchase the car outright or lease the car. Purchasing provides more certainty about the expense over a period of time, but leasing provides more flexibility. The type of car you buy depends on your personal preference and how you intend to use it (similar to whether you are buying a base load plant or a peaking plant). A sports car is great for a small segment of the population but likely does not work for those with a family. Similar to the criteria for energy resource decisions, the choice you make is based on what you think is important and what you think will happen in the future. This is a judgment call, and different people will come to different conclusions.

The term "reasonable" is included in the phrase, "lowest reasonable system cost" to indicate there can be different viewpoints on what constitutes lowest cost. In addition to the tradeoff between upfront and operating costs, a big variable for the production of electricity is fuel. If the utility wants the cheapest fuel price, fuel contracts tend to be for a short period of time. If the utility wants a more stable fuel price, fuel contracts tend to be for a longer period of time and typically involve a premium over current short-term prices. This is very similar to the tradeoff between a fixed-rate mortgage and a variable-rate mortgage.

Another way to view the "lowest reasonable system cost" issue is to separate those costs paid as part of the purchase price from those costs not allocated to any one party. A good example is air emissions. Most people would agree that emissions have a cost, but it is not a cost allocated to any one party. Instead, people pay for air

emissions based upon their future impacts on health and aesthetics.

A third way to think about costs is to recognize differences in reliability or control over the generating resource. Some would argue that the greatest control and reliability comes from building a traditional generating resource close to the area where the power will be consumed. They might argue that generation resources produced by efficiency programs or purchased power contracts produce less reliable power, or power for which there is less control. Advocates for efficiency programs or purchasing power would say such generation sources have reliability and control equivalent to the rest of the system. The need for super-reliability or immediate control is not worth the price of the tradeoffs, they would argue. Once again, this is a judgment call where opinions can differ. We need to know how you would make the judgment.

Of The Options For New Resources Available, Which Ones Do You Prefer?

Another way to think about future generation resources is to consider the types of projects themselves. Each has advantages and disadvantages. As part of this Deliberative Poll™, you will discuss four broad options.

One option for meeting the need for additional generation is to use existing generation more efficiently. Most of these changes take place on the customer's side of the meter. Energy efficiency services and technologies produce a number of small savings. Averaged together, these can produce a large block of energy resources.

The most traditional means of filling a need would be to build a new power plant, likely a natural gas plant. If the needs of several utility companies were merged, the answer might be a coal plant. This traditional approach has advantages and disadvantages.

A third way to meet the need for new generation resources is with renewable energy. The most cost-effective current sources of renewable energy are wind power or methane gas from landfills. While renewable energy is currently more expensive to build than traditional generation,

the prices are coming down. The fuel is either free or very inexpensive.

A fourth way to meet the need is through purchasing power from the marketplace. This option is not exactly parallel with the other three options. Purchased power could come from traditional sources, renewable sources, or efficiency programs. The key is that the price is determined by supply and demand in the marketplace at the time the Company is ready to commit to a purchase.

If Some Of The Options Cost A Little More - Would You Still Be In Favor? If You Are Still In Favor, How Much More Would You Be Willing To Pay?

This view of the issues combines the previous two discussions on the page. It combines the view of what constitutes "lowest reasonable system cost" with your preferences for a particular resource. If all options and resources cost the same and have the same or similar reliability, then you may have a preference for a particular solution. But, all options do not cost the same. Do you still favor a particular solution and are you willing to pay more for that solution? It may be helpful to answer within a context. It may be helpful to think of your monthly energy bill and to consider how much more you would spend.

What Is Meant By "Pay More" Under A Rate Freeze

In order to assess the value that the customer would put on a service or program, participants in a Deliberative Polling™ such as this are typically asked how much more they would pay for a certain service and/or resource. That question is being asked here. However, as a result of El Paso Electric's bankruptcy plan and settlement approved by the PUCT, which includes a 10-year base rate freeze (fuel costs are not frozen) your base rates will not go up.

It might be a little confusing, but think in terms of "less savings" when you are asked if you would pay more for a particular choice. The words "would you pay more", mean one generation resource compared to another. Your rates should not go up.

There Are No Right or Wrong Answers

We Need To Know Your Opinions About Low-Income Programs.

The El Paso Electric service territory has a significant population of customers who are at or below national poverty level standards. State law requires that utility companies offer energy efficiency programs equitably. In designing programs for customers, especially energy efficiency programs, it is possible to target those programs towards a particular group. For example, it would be possible to target a larger percentage of energy efficiency dollars towards those who are in greater financial need. The issue here concerns your ideas about equity. Once again, this is a judgment call, and your definition of fair treatment and the best use of limited funds may differ from those of your neighbors.

Are You Interested In A Renewable Energy Choice Program?

The issue here is individual choice programs. This choice affects only the customer making the commitment. These programs are called green choice or green marketing programs. They are similar to the choice you make when you buy recycled products. We need to know if you are interested in an individual choice program and, if so, at what dollar level.

Some customers believe the utility should find more environmentally friendly sources of generation and are willing to pay more for those sources. They see this as a community decision with the costs spread to all customers. That is a separate issue.

Are You Interested In New Services?

The fundamental nature of electric utilities is changing. Over the next few years, new services will be available both from El Paso Electric and from competing companies. Some of these services may include things such as energy efficient mortgages, loans for energy efficient products and service warranties on appliances. Other new options might include the way electricity is billed so that electricity is cheaper at times when demand is low and more expensive at times when demand is high. Another billing option might include a credit on your bill if the Company can interrupt use at periods of highest demand.

These options will be new, and you will have a choice. We need to know how you feel about these potential choices.

What Do You Think About El Paso Electric And Its Services?

You will be asked some questions about services you currently receive from El Paso Electric. These include the length of time it takes to restore electricity when there is a power outage, the frequency of power outages, and having enough electricity for the hottest days. We need to know whether the assistance you received from the Company was helpful and courteous, and what you think about the cost of electricity. We need to know if power surges have been a problem and how you compare electric service to some other common forms of service.

What Do You Think About This Deliberative Polling™ Process?

Deliberative Polling™ is a new way to gather informed customer opinions. By the end of the 1-1/2 days, you will have invested a significant amount of time as a participant. This is only the ninth Deliberative Poll™ in the world, so your opinion is useful in making improvements.

We will want to know how you rate the process and if you believe it was fair and objective. We will also give you an opportunity to write open-ended comments concerning any aspect of the process.

Overview of the Deliberative Poll™

Professor James Fishkin, Chairman of the Government Department at the University of Texas at Austin, developed the concept of Deliberative Polling™ as an approach for determining informed public opinion. The approach received international attention during the National Issues Convention held in Austin in January 1996.

Deliberative Polling™ differs significantly from ordinary polls and surveys. As indicated in Professor Fishkin's book, *The Voice of the People, Public Opinion & Democracy*, ordinary polls and surveys examine what the public is thinking about different issues at a particular time. Deliberative

Polling™ attempts to determine what the public would think about different issues if given an opportunity to learn more about all sides of the issues.

First, a poll is taken that determines public opinions about the issues under consideration. Then a scientifically random sample of the population is identified. That group meets in one location where they are immersed in the issues through carefully balanced briefing materials, intensive discussions in small groups and the opportunity to listen to and question experts with differing views.

At the end of the process, the group is again surveyed to determine whether their views on the various issues changed and to what extent. The resulting survey offers a representation of the considered judgments of the entire population from which the sample was drawn.

This process is vital to El Paso Electric in that it provides important information about what values customers place on certain approaches or combinations of approaches for meeting future energy needs and for designing future energy services. The final plan that El Paso Electric submits to the PUCT will affect customers for many years.

Role Of The Moderators And Facilitators

Each small group has a trained professional moderator. Their job is to help you walk through the costs and consequences of the various options, keep the discussion on track and make sure all participants have a chance to ask their questions. Moderators are recruited from outside El Paso Electric in order to maintain neutrality during the process. In addition, each group has an El Paso Electric volunteer facilitator. Their job is to solve problems.

Ron Lehr, a former chairman of the Colorado Public Utility Commission, will moderate the large group sessions.

A Special Thanks To:

- The University of Texas at El Paso
- KCOS – Channel 13
- The University of Texas at Austin
- Deliberative Polling™ Advisory Group
- City of El Paso
- Public Utility Commission of Texas
- New Mexico Public Utility Commission
- Speakers and Panel Members

Friday, August 15		Saturday, August 16	
12:30 – 1:30	Registration and Refreshments (light snack) meet small group members	7:00 – 8:00	Breakfast
1:30 – 2:15	Welcome and Orientation	8:00 – 9:00	Small Group Session #3: Efficiency and Renewables
2:15 – 3:15	Small Group Session #1: What Factors are Important in Planning for Future Energy Needs	9:00 – 10:00	Small Group Session #4: Competition and the Changing Nature of Utility Services
3:15 – 4:15	Small Group Session #2: Traditional Generation and Purchased Power	10:00 – 10:15	Break
4:15 – 4:30	Break (refreshments and cookies)	10:15 – 11:45	Large Group Session #2: Experts and Advocates Answer Questions
4:30 – 6:00	Large Group Session #1: Experts and Advocates Answer Questions	11:45 – 12:15	Small Group Session #5: Short Session to Select Questions for Regulatory Panel
6:00 – 7:00	Reception	12:15 – 1:15	Lunch (box lunch with your group)
		1:15 – 2:45	Large Group Session #3: View and Plans of Regulators (taped for broadcast)
		2:45 – 3:00	Break
		3:00 – 4:00	Small Group Session #6: This is What We Need to Know—What is Important to Customers
		4:00 – 5:30	Large Group Session #4: Panel of Interested Parties
		5:30 – 6:00	Closing and Thanks
		6:00 – 6:30	Complete Questionnaire and Receive Check

